

INRE Project — Progress Report, Assumptions, Results & Next Steps

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Author context: Energy systems modelling study on the Germany electricity grid using PyPSA-Eur with an INRE (Integrated Nuclear & Renewable Energy) extension layer.

Status: Phase 2 pipeline **implemented and executed** — all five scenarios solved successfully on 22 June 2026.

Executive Summary

This project builds a custom modelling layer on top of [PyPSA-Eur](#) to answer one question:

During a Dunkelflaute (extended low wind/solar) event in winter, can new nuclear technologies (SMR, MSR, LFR) cost-effectively complement a high-renewable German electricity system?

What has been completed:

- 1. A full Snakemake-integrated INRE workflow (configs, data, Python scripts, rules).
- 2. A five-scenario matrix: base reference, Dunkelflaute stress, and three Dunkelflaute + nuclear technology variants.
- 3. Successful execution of all five linear programming (LP) optimisations using the HiGHS solver.
- 4. A cross-scenario comparison script producing KPI tables ([results/inre-comparison/](#)).

Headline preliminary findings:

Finding	Engineering interpretation
Dunkelflaute stress raises operating cost by +69 M EUR (+25.6%) over the 2-week window	VRE shortfall is met primarily by CCGT (+247%) and slightly more coal/lignite , not by storage or nuclear
Wind capacity factor drops ~19% , solar ~37% under stress	The stress model works as intended; worst days were auto-detected inside the simulation window
New nuclear builds ~0 MW (numerical dust only: 0.0001–0.002 MW total)	At placeholder 2050 costs (~18–22 kEUR/MW/year annuitised CAPEX) and a 2-week optimisation window , nuclear cannot recover investment; gas is cheaper
CO ₂ limit is not binding (3.3 Mt emitted vs 19.2 Mt allowed over 2 weeks)	The 500 Mt/year cap is a loose placeholder; coal/lignite remain in the dispatch stack
Average load ~63 GW , peak ~76 GW	Consistent with a winter week in a 2050-horizon, electricity-only DE model

Critical caveat: Results are **preliminary and exploratory**. Many inputs are explicit placeholders ([INRE assumption](#)). The 2-week / 3-hour snapshot design prioritises fast iteration over annual energy balance or investment realism.

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1. Project Objective & Scope

1.1 Research question

Germany has phased out commercial nuclear generation (last plants shut down in April 2023). As the share of wind and solar grows, **Dunkelflaute** periods — multi-day episodes of simultaneously low wind and solar output, typically in winter — create reliability and adequacy challenges.

The INRE project uses PyPSA-Eur to simulate:

- A **2050-technology-cost, 2024-renewable-capacity** vision of the German grid.
- A **deliberate VRE availability stress** representing Dunkelflaute.
- Optional **new build** of advanced nuclear at former reactor sites (SMR, MSR, LFR).

1.2 Scope boundaries (what the model does *not* include)

Excluded	Implication
Sector coupling (heat, transport, industry)	Electricity-only; no heat pumps, EV load, or H ₂ demand from other sectors
Cross-border trading detail	Germany-only (<code>countries: [DE]</code>); neighbouring countries not modelled
Existing nuclear	Correctly zero — all German NPPs filtered out / shut down
Unit commitment / start-up costs	LP dispatch with ramp limits only; no binary on/off
Perfect foresight over full year	Single 2-week window; investment decisions based on partial year exposure

2. What Was Built — Technical Inventory

2.1 Repository modifications

The INRE layer was added **without forking** PyPSA-Eur core logic. Changes:

Component	Path	Purpose
Base config	<code>config/inre/config.base.yaml</code>	Germany, Jan 2021 window, 10 clusters, 3h resolution
Multi-scenario driver	<code>config/inre/config.scenarios.yaml</code>	Runs all scenarios with shared build resources
Scenario overrides	<code>config/inre/scenarios.yaml</code>	Per-scenario Dunkelflaute / nuclear flags
Phase 1 fast-dev config	<code>config/inre/config.phase1-fast.yaml</code>	March 2013, tutorial mode, smoke tests
Dunkelflaute parameters	<code>data/inre/dunkelflaute.yaml</code>	Stress factors, auto worst-day selection
Nuclear costs (placeholder)	<code>data/inre/custom_costs_nuclear.csv</code>	SMR / MSR / LFR techno-economic data
Candidate sites	<code>data/inre/custom_powerplants_nuclear_DE.csv</code>	5 former NPP locations
Dunkelflaute script	<code>scripts/inre/apply_dunkelflaute.py</code>	VRE profile derating
Nuclear script	<code>scripts/inre/add_nuclear_technologies.py</code>	Extendable generators at sites
Orchestrator	<code>scripts/inre/apply_inre_network.py</code>	Snakemake entry point
Comparison tool	<code>scripts/inre/compare_scenarios.py</code>	KPI tables and charts
Snakemake rules	<code>rules/inre.smk</code>	<code>apply_inre_network</code> rule; solve input routing
Snakefile include	Snakefile line 86	<code>include: "rules/inre.smk"</code>
Solve routing patch	<code>rules/solve_electricity.smk</code>	<code>_input_solve_network</code> uses INRE network when needed
Documentation	<code>INRE-README.md</code>	Operational reference

2.2 Workflow architecture

```
retrieve → build base network → cluster (10 nodes) → renewable profiles
  → add_electricity → prepare_network
    → [INRE: apply_inre_network] ← only for Dunkelflaute / nuclear scenarios
      → solve_network (HiGHS LP)
        → results/<scenario>/networks/*.nc
          → compare_scenarios.py
```

The INRE step sits **between** `prepare_network` and `solve_network`. For the `base` scenario (no Dunkelflaute, no nuclear), the prepared network is passed directly to the solver.

2.3 Execution record (22 June 2026)

Command executed (from terminal log):

```
snakemake --call solve_elec_networks \
  --configfile config/inre/config.base.yaml \
  --configfile config/inre/config.scenarios.yaml
```

Metric	Value
Scenarios solved	5 / 5
Solver	HiGHS (interior-point)
Base case solve time	~306 s (~5.1 min)
Dunkelflaute scenarios	~375–382 s (~6.2–6.4 min)

Metric	Value
Peak memory	~1.9 GB
Snapshots per scenario	112 (14 days × 8 blocks/day at 3h resolution)
Optimality	HiGHS reported "Unknown" status due to small primal-dual gap (~3×10 ⁻⁶ relative); solutions parsed successfully

Output networks:

```
results/base/networks/base_s_10_elec_.nc
results/dunkelflaute/networks/base_s_10_elec_.nc
results/dunkelflaute-smr/networks/base_s_10_elec_.nc
results/dunkelflaute-msr/networks/base_s_10_elec_.nc
results/dunkelflaute-lfr/networks/base_s_10_elec_.nc
```

Comparison table written to `results/inre-comparison/comparison_table.csv`.

3. Modelling Methodology & Calculation Logic

3.1 PyPSA-Eur base model

PyPSA-Eur constructs a **linear optimal power flow / capacity expansion** model:

- **Sets:** buses (10 clustered regions), generators (~5,876 units), lines (22), loads, storage (battery, H₂).
- **Decision variables:** generator dispatch `p[g,t]`, extendable capacity `p_nom[g]`, storage operation, line flows.
- **Objective:** minimise total system cost = annuitised CAPEX + fixed O&M + variable O&M + fuel costs, weighted by snapshot duration.
- **Constraints:** nodal power balance (Kirchhoff), generator ramp limits, line capacities, CO₂ cap, renewable potentials.

The model is a **single-year LP** with snapshot weighting to represent annual costs from a subset of timesteps.

3.2 Spatial clustering — 10 nodes

Germany's transmission network is aggregated to **10 zones** using PyPSA-Eur's `cluster_network` rule. Each zone has:

- Aggregated demand (ENTSO-E load, distributed to buses).
- Renewable generators (potentials from Atlite weather × installable capacity).
- Conventional plants from powerplantmatching, mapped to nearest bus.
- Inter-zonal AC lines (22 remaining after clustering).

Engineering note: 10 clusters capture north–south wind/solar heterogeneity at coarse resolution. Dunkelflaute is applied **nationally** (same derating factor at all nodes simultaneously), which is conservative for a spatially correlated winter anticyclone but ignores regional differences in cloud cover.

3.3 Temporal setup — why 2 weeks at 3-hour resolution?

Parameter	Value	Rationale
<code>snapshots.start</code>	2021-01-25	Start of documented Jan 2021 cold/Dunkelflaute episode
<code>snapshots.end</code>	2021-02-08	14-day window (end exclusive in PyPSA → 13 full days + partial)
<code>resolution_elec</code>	3h	112 snapshots; target ≤ 15 min solve per scenario
Weather cutout	<code>europe-2021-sarah3-era5</code>	ERA5 + SARA-3 satellite irradiance

Snapshot weighting: Each snapshot represents 3 hours. PyPSA assigns `snapshot_weightings.objective` so that costs integrated over all snapshots correspond to an **annualised** objective. For a 2-week window:

$$\text{period fraction of year} = \frac{\sum w_t}{8760} \approx 0.0384 \text{ ; (3.84\%)}$$

Therefore, the reported **objective** (~270 M EUR for base) is the **cost attributed to this 2-week slice**, not the full annual system cost. A rough annualisation (base case):

$$\text{Annualised OPEX} \approx \frac{269.7 \text{ M EUR}}{0.0384} \approx 7.0 \text{ bn EUR/year}$$

This annualisation is **not rigorous** for investment decisions because capacity expansion sees only 2 weeks of operational patterns.

3.4 Dunkelflaute implementation logic

Implemented in `scripts/inre/apply_dunkelflaute.py`. Operates on `generators_t.p_max_pu` (per-unit availability) before the solve.

Step 1 — Identify stress window:

Two modes (configured in `data/inre/dunkelflaute.yaml`):

Mode	Config key	Behaviour
Auto worst days	<code>auto_worst_days: 5</code>	Compute daily national VRE score; pick 5 lowest days

Mode	Config key	Behaviour
Fixed calendar	<code>time_start / time_end</code>	Use explicit date range

VRE score formula (lower = worse Dunkelflaute):

For each snapshot t and day d :

$$d = \frac{1}{T_d} \left[\sum_{g \in T_d} p_{\text{max_pu}}(g,t) \cdot p_{\text{nom_g}} + \sum_{g \in \text{solar}} p_{\text{max_pu}}(g,t) \cdot p_{\text{nom_g}} \right]$$

Capacity-weighted average availability. The 5 days with the lowest score are selected.

Auto-selected worst days (this run):

Date	Notes
2021-01-25	First day of simulation window; deep winter, low solar
2021-01-28	Mid-period low VRE
2021-01-30	Weekend low wind
2021-01-31	Continuation
2021-02-02	Late-window stress day

Step 2 — Apply derating factors:

Inside the stress mask:

$$p_{\text{max_pu}}(g,t) \rightarrow p_{\text{max_pu}}(g,t) \cdot f_{\text{carrier}}$$

Carrier group	Factor	Retained availability
Wind (onshore, offshore AC/DC/float)	0.15	15%
Solar (fixed, tracking)	0.10	10%

Step 3 — Edge ramping:

Over `ramp_hours: 6` snapshots (18 hours at 3h resolution), availability linearly ramps between full and stressed values at the window boundaries. This avoids a discontinuous step change that would create artificial ramping infeasibilities.

Physical interpretation: This is a **scenario stress test**, not a meteorological reanalysis product. Factors 0.15/0.10 represent an extreme but plausible simultaneous collapse of wind and solar capacity factors — consistent with a stagnant high-pressure system over Central Europe in January.

3.5 Nuclear technology addition logic

Implemented in `scripts/inre/add_nuclear_technologies.py`.

For each row in `custom_powerplants_nuclear_DE.csv` matching the scenario carrier:

1. Find **nearest network bus** by Euclidean distance in lat/lon.
2. Add an **extendable generator** with:
 - `p_nom = 0, p_nom_extendable = True`
 - `p_nom_max = 1500 MW` (config default)
 - Costs from processed `costs_2050_processed.csv` (annuitised from `custom_costs_nuclear.csv`)
 - `p_max_pu = 0.9` (90% availability factor)
 - `p_min_pu = 0.3` (30% minimum stable load)
 - `ramp_limit_up/down = 0.5 p.u./hour`

The optimiser then decides `p_nom_opt` balancing CAPEX against dispatch savings over the snapshot window.

Annuitised capital cost example (SMR):

From processed costs: `capital_cost = 18,335 EUR/MW` (per MW of installed capacity, annualised over 60-year lifetime at 2050 discount rate from technology-data).

With marginal cost ~12 EUR/MWh (fuel + VOM + FOM), nuclear is **cheap to operate** but **expensive to build** — classic baseload economics.

3.6 CO₂ constraint

```
electricity:
  co2limit_enable: true
  co2limit: 500.e+6 # tonnes CO2 per year
```

PyPSA scales this to the snapshot period:

$$\text{CO2Limit}_{\text{period}} = 500 \times 10^6 \times \frac{336}{8760} \approx 19.18 \text{ Mt}$$

Actual emissions (2-week window):

Scenario	CO ₂ (Mt)	Limit (Mt)	Utilisation
base	3.28	19.18	17%
dunkelflaute	3.85	19.18	20%
dunkelflaute-smr	3.85	19.18	20%

The constraint is **non-binding**. Coal and lignite dispatch freely up to their installed capacity. This is a major policy assumption gap for a 2050 decarbonisation study.

3.7 Key performance indicators (KPIs)

KPI	Calculation in PyPSA
Operating cost (OPEX)	<code>n.statistics.opex()</code> — variable + fixed O&M over weighted snapshots
Investment (CAPEX)	<code>n.statistics.capex()</code> — annuitised capital for optimal capacity
Total objective	<code>n.objective</code> — LP objective value
Generation by carrier	<code>n.generators_t.p</code> × snapshot weights, summed
Optimal capacity	<code>n.generators.p_nom_opt</code> for extendable units
CO ₂ emissions	$\sum_g p_{g,t} \cdot w_t \cdot \text{co2_emissions}_g$

4. Complete List of Manual Assumptions

Every assumption below is either explicitly tagged **INRE assumption** in data files or implied by config choices. **None should be used for publication without review.**

4.1 Geographic & temporal scope

#	Assumption	Value	Source / tag	Sensitivity
T1	Modelled country	Germany only (DE)	Config choice	High — excludes imports
T2	Spatial resolution	10 clusters	Config choice	Medium
T3	Simulation window	2021-01-25 → 2021-02-08	Historical Dunkelflaute event	High
T4	Temporal resolution	3 hours (112 snapshots)	Runtime constraint	Medium–High
T5	Planning horizon for costs	2050	PyPSA technology-data	High
T6	Renewable fleet year	2024 capacities	powerplantmatching	Medium

4.2 Dunkelflaute stress parameters

#	Assumption	Value	Source / tag	Sensitivity
D1	Wind availability during stress	15% of normal (<code>wind_factor: 0.15</code>)	INRE manual	Very high
D2	Solar availability during stress	10% of normal (<code>solar_factor: 0.10</code>)	INRE manual	Very high
D3	Stress duration	5 auto-selected worst days	INRE manual	High
D4	Edge ramp	6 snapshots (18 h)	INRE manual	Low–Medium
D5	Spatial correlation	Uniform national derating	INRE simplification	Medium
D6	Fallback fixed window	2021-01-28 → 2021-02-03	Not used (auto mode active)	—

4.3 Nuclear technology placeholders

#	Parameter	SMR	MSR	LFR	Unit	Tag
N1	Investment cost	4500	5200	4800	EUR/kW	INRE assumption
N2	Fixed O&M	3.5	4.0	3.8	%/year	INRE assumption
N3	Variable O&M	3.0	3.5	3.2	EUR/MWh	INRE assumption
N4	Fuel cost	3.0	2.5	2.8	EUR/MWh _{th}	INRE uranium estimate
N5	Electrical efficiency	0.33	0.35	0.34	p.u.	INRE assumption
N6	Lifetime	60	50	55	years	INRE assumption
N7	CO ₂ intensity	0	0	0	t/MWh	INRE assumption

#	Parameter	SMR	MSR	LFR	Unit	Tag
N8	Availability (p_max_pu)	0.90	0.90	0.90	p.u.	Script default
N9	Minimum load (p_min_pu)	0.30	0.30	0.30	p.u.	Script default
N10	Ramp rate	0.50	0.50	0.50	p.u./hour	Script default
N11	Max build per site	1500	1500	1500	MW	Config default
N12	National build cap	None	None	None	GW	Not implemented
N13	Candidate sites	5 former NPP locations	Manual CSV	Medium		
N14	Commissioning year	2045 (DateIn)	Manual CSV	Low (not enforced in LP)		

Processed annuitised costs (from solver input):

Carrier	Capital cost (EUR/MW/a)	Marginal cost (EUR/MWh)
nuclear-smr	18,335	12.10
nuclear-msr	22,430	10.64
nuclear-lfr	20,204	11.44
CCGT (reference)	4,849	43.27
onwind (reference)	4,557	0.01
solar (reference)	1,415	0.01

4.4 Policy & fleet assumptions

#	Assumption	Value	Implication
P1	CO ₂ cap	500 Mt/year	Non-binding; coal/lignite remain economic
P2	Existing nuclear	Zero capacity	Correct for post-2023 Germany
P3	Coal/lignite retirement	Filter keeps plants with DateOut > 2025	~40 GW coal still in fleet
P4	Extendable technologies	VRE, OCGT, CCGT, battery, H ₂	Gas and storage can expand
P5	Sector coupling	Disabled	No flexibility from heat/transport
P6	Dynamic line rating	Disabled	Fixed thermal limits
P7	Offshore depth limits	Disabled (max_depth: false)	Unrestricted offshore siting

4.5 Solver & numerical

#	Assumption	Value
S1	Solver	HIGHS (free LP)
S2	Optimality tolerance	Default; P-D gap $\sim 3 \times 10^{-6}$ triggers "Unknown" status
S3	Foresight	Perfect within 2-week window
S4	Storage	Battery/H ₂ nearly unused (see results)

5. Input Data Sources vs Assumptions

Data category	Source	Free?	INRE override?
Transmission grid topology	OpenStreetMap (PyPSA archive)	Yes	No
Power plant database	powerplantmatching	Yes	Filter for DE, retirement dates
Renewable capacities (2024)	powerplantmatching estimate	Yes	No
Electricity demand	ENTSO-E Transparency Platform	Yes	No
Weather (wind speed, irradiance)	ERA5 + SARA-H-3 via Atlite	Yes	Dunkelflaute derating applied
Technology costs (2050)	PyPSA technology-data	Yes	Nuclear costs overridden
Nuclear availability (legacy)	IAEA PRIS	Yes	Not used (no existing nuclear)
Nuclear new-build costs	Manual placeholder	—	Yes — must replace
Dunkelflaute severity	Manual scenario parameter	—	Yes — calibrate to events
Reactor siting	Manual CSV at former sites	—	Yes — policy-driven

6. Scenario Matrix & Cross-Scenario Comparison

This section is the **master reference** for what each scenario changes relative to the others: config overrides, data files, pipeline steps, active assumptions, and measured outcomes.

6.1 Scenario overview

Scenario ID	Run name	Results folder	Purpose
base	inre-de-base	results/base/	Reference: normal Jan 2021 VRE profiles, no stress, no new nuclear
dunkelflaute	inre-de-dunkelflaute	results/dunkelflaute/	Dunkelflaute stress only — quantify adequacy gap
dunkelflaute-smr	inre-de-df-smr	results/dunkelflaute-smr/	Stress + Small Modular Reactor as extendable option
dunkelflaute-msr	inre-de-df-msr	results/dunkelflaute-msr/	Stress + Molten Salt Reactor as extendable option
dunkelflaute-lfr	inre-de-df-lfr	results/dunkelflaute-lfr/	Stress + Lead-cooled Fast Reactor as extendable option

All scenarios inherit from `config/inre/config.base.yaml` + `config/inre/config.scenarios.yaml`, with per-scenario overrides in `config/inre/scenarios.yaml`.

Shared build resources (`shared_resources.policy: true`) mean grid topology, clustering, weather cutout, and the prepared network are built **once** and reused. Only the INRE modification step (`apply_inre_network`) and the LP solve differ per scenario.

6.2 Shared baseline — identical across all five scenarios

These settings are **not changed** between scenarios. They form the common modelling foundation.

Category	Parameter	Value	Config / data source
Geography	Country	Germany only (DE)	<code>config.base.yaml</code>
Spatial	Clusters	10 nodes	<code>scenario.clusters: [10]</code>
Temporal	Snapshot window	2021-01-25 → 2021-02-08 (14 days)	<code>snapshots</code>
Temporal	Resolution	3-hour (112 snapshots)	<code>clustering.temporal.resolution_elec: 3h</code>
Weather	Cutout	<code>europe-2021-sarah3-era5</code>	<code>atlite.default_cutout</code>
Costs	Planning year	2050	<code>costs.year: 2050</code>
Fleet	Renewable capacity year	2024 (powerplantmatching)	<code>estimate_renewable_capacities.year</code>
Fleet	Existing nuclear	0 MW (all shut down)	<code>powerplants_filter</code>
Fleet	Coal + lignite	~40 GW (plants with <code>DateOut > 2025</code>)	<code>powerplants_filter</code>
Policy	CO ₂ cap	500 Mt/year (non-binding in results)	<code>electricity.co2limit: 500.e+6</code>
Extendable	Generators	solar, solar-hsat, onwind, offwind-*, OCGT, CCGT	<code>extendable_carriers.Generator</code>
Extendable	Storage	battery, H ₂ store, H ₂ pipeline	<code>extendable_carriers</code>
Solver	Engine	HiGHS (free LP)	<code>solving.solver.name: highs</code>
INRE	Master switch	<code>inre.enabled: true</code>	All scenarios

6.3 Configuration delta matrix — what changes per scenario

Legend:  = active / added · — = same as base ·  = off / not present

Setting	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Dunkelflaute stress					
<code>inre.dunkelflaute.enabled</code>	false	true	true	true	
Dunkelflaute config file	—	<code>data/inre/dunkelflaute.yaml</code>	same	same	
Wind derating factor	—	0.15 (15%)	same	same	
Solar derating factor	—	0.10 (10%)	same	same	
Stress day selection	—	<code>auto_worst_days: 5</code>	same	same	
Edge ramp	—	<code>ramp_hours: 6</code>	same	same	
New nuclear build			 SMR	 MSR	
<code>inre.nuclear.extendable_carriers</code>	[]	[]	[nuclear-smr]	[nuclear-msr]	

Setting	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	c
Nuclear sites file	—	—	<code>custom_powerplants_nuclear_DE.csv</code>	same	
Max build per site	—	—	1500 MW	1500 MW	
<code>costs.custom_cost_fn</code> (explicit)	inherited	inherited	✓ explicit	✓ explicit	
<code>pypsa_eur.Generator</code> carrier list	default	default	+ <code>nuclear-smr</code>	+ <code>nuclear-msr</code>	
Pipeline: <code>apply_inre_network</code>	Skipped (copy)	Dunkelflaute only	Dunkelflaute + nuclear	Dunkelflaute + nuclear	1
Output network suffix	<code>_elec_.nc</code>	<code>_elec_.nc</code>	<code>_elec_.nc</code>	<code>_elec_.nc</code>	
INRE intermediate file	—	<code>*_inre.nc</code>	<code>*_inre.nc</code>	<code>*_inre.nc</code>	

6.4 Scenario-by-scenario — what was added or changed vs `base`

6.4.1 `base` (reference)

Aspect	Detail
Changes vs PyPSA-Eur default	Germany-only filter, Jan 2021 window, 10 clusters, 3h resolution, INRE config block present but inactive
INRE modifications	None — prepared network copied directly to solver
Added generators	None
Modified time series	None — raw Atlite VRE profiles
Active INRE assumptions	T1–T6, P1–P7, S1–S4 only (see Section 4)
Purpose	Baseline dispatch and cost under normal winter weather

6.4.2 `dunkelflaute` (+1 change vs `base`)

Aspect	Detail
Added vs <code>base</code>	Dunkelflaute VRE derating via <code>apply_dunkelflaute.py</code>
Modified time series	<code>generators_t.p_max_pu</code> for all wind and solar generators on 5 worst VRE days
Auto-selected stress days	2021-01-25, 2021-01-28, 2021-01-30, 2021-01-31, 2021-02-02
Added generators	None
Removed / disabled	Nothing
Active INRE assumptions	Base assumptions + D1–D6 (Dunkelflaute parameters)
Script / rule invoked	<code>rules/inre.smk</code> → <code>apply_inre_network</code> → <code>apply_dunkelflaute.py</code>
Purpose	Isolate the cost and dispatch impact of a Dunkelflaute event without nuclear

6.4.3 `dunkelflaute-smr` (+2 changes vs `base`)

Aspect	Detail
Added vs <code>base</code>	(1) Dunkelflaute derating — same as <code>dunkelflaute</code> ; (2) five extendable SMR generators
Added generators	5 × <code>nuclear-smr</code> at former NPP sites (Grohnde, Brokdorf, Isar, Emsland, Neckarwestheim)
Added carrier	<code>nuclear-smr</code> registered in <code>pypsa_eur.Generator</code> and <code>custom_costs_nuclear.csv</code>
Generator parameters	<code>p_nom_extendable=True</code> , <code>p_nom_max=1500 MW</code> , <code>p_max_pu=0.9</code> , <code>p_min_pu=0.3</code> , <code>ramp=0.5 p.u./h</code>
Cost assumptions used	Investment 4500 EUR/kW, FOM 3.5%/a, VOM 3 EUR/MWh, fuel 3 EUR/MWh _{th} , $\eta=0.33$, lifetime 60 y
Active INRE assumptions	D1–D6 + N1–N14 (SMR column)
Script / rule invoked	<code>apply_dunkelflaute.py</code> then <code>add_nuclear_technologies.py</code>
Purpose	Test whether SMR can economically cover the Dunkelflaute gap

6.4.4 `dunkelflaute-msr` (+2 changes vs `base`)

Aspect	Detail
Added vs <code>base</code>	(1) Dunkelflaute derating; (2) three extendable MSR generators
Added generators	3 × <code>nuclear-msr</code> at Grohnde, Brokdorf, Isar

Aspect	Detail
Added carrier	nuclear-msr
Cost assumptions used	Investment 5200 EUR/kW, FOM 4.0%/a, VOM 3.5 EUR/MWh, fuel 2.5 EUR/MWh_th, η =0.35, lifetime 50 y
Annuitised CAPEX	22,430 EUR/MW/a (highest of three nuclear options)
Marginal cost	10.64 EUR/MWh (lowest operating cost of three)
Active INRE assumptions	D1–D6 + N1–N14 (MSR column)
Purpose	Test molten salt reactor under same Dunkelflaute stress

6.4.5 dunkelflaute-lfr (+2 changes vs base)

Aspect	Detail
Added vs base	(1) Dunkelflaute derating; (2) three extendable LFR generators
Added generators	3 × nuclear-lfr at Grohnde, Brokdorf, Emsland
Added carrier	nuclear-lfr
Cost assumptions used	Investment 4800 EUR/kW, FOM 3.8%/a, VOM 3.2 EUR/MWh, fuel 2.8 EUR/MWh_th, η =0.34, lifetime 55 y
Annuitised CAPEX	20,204 EUR/MW/a
Marginal cost	11.44 EUR/MWh
Active INRE assumptions	D1–D6 + N1–N14 (LFR column)
Purpose	Test lead-cooled fast reactor under same Dunkelflaute stress

6.5 Nuclear candidate sites added per scenario

Rows from `data/inre/custom_powerplants_nuclear_DE.csv`. Only sites matching the scenario carrier are instantiated.

Site (former NPP)	Lat	Lon	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Grohnde	51.906	9.401	✓ SMR	✓ MSR	✓ LFR
Brokdorf	53.851	9.345	✓ SMR	✓ MSR	✓ LFR
Isar	48.617	12.293	✓ SMR	✓ MSR	✗
Emsland	52.471	7.321	✓ SMR	✗	✓ LFR
Neckarwestheim	49.040	9.175	✓ SMR	✗	✗
Total candidate generators			5	3	3
Max theoretical build			7.5 GW	4.5 GW	4.5 GW

Each generator is mapped to the nearest PyPSA bus via Euclidean distance in geographic coordinates.

6.6 Assumptions active per scenario

Which assumption groups from Section 4 apply to each scenario:

Assumption group	ID range	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Geographic & temporal scope	T1–T6	✓	✓	✓	✓	✓
Dunkelflaute stress	D1–D6	✗	✓	✓	✓	✓
Nuclear SMR economics	N1–N14 (SMR)	✗	✗	✓	✗	✗
Nuclear MSR economics	N1–N14 (MSR)	✗	✗	✗	✓	✗
Nuclear LFR economics	N1–N14 (LFR)	✗	✗	✗	✗	✓
Policy & fleet	P1–P7	✓	✓	✓	✓	✓
Solver & numerical	S1–S4	✓	✓	✓	✓	✓

Nuclear technology cost comparison (assumptions only active in respective scenario):

Parameter	SMR	MSR	LFR	Unit
Overnight investment	4,500	5,200	4,800	EUR/kW
Annuitised capital cost	18,335	22,430	20,204	EUR/MW/a
Marginal (operating) cost	12.10	10.64	11.44	EUR/MWh

Parameter	SMR	MSR	LFR	Unit
Electrical efficiency	0.33	0.35	0.34	p.u.
Lifetime	60	50	55	years
Availability factor	0.90	0.90	0.90	p.u.
Min stable load	0.30	0.30	0.30	p.u.
Candidate sites	5	3	3	count
Source tag	INRE assumption	INRE assumption	INRE assumption	—

6.7 Pipeline modification summary

What happens in the Snakemake workflow for each scenario after the shared build phase:

Pipeline step	base	dunkelflaute	dunkelflaute-smr / msr
prepare_network	✔ shared	✔ shared	✔ shared
apply_inre_network	⏸ skipped (file copy)	✔ apply_dunkelflaute	✔ apply_dunkelflaute add_nuclear_technology
Input to solve_network	resources/.../base_s_10_elec.nc	results/.../base_s_10_elec__inre.nc	results/.../base_s_10_elec__inre.nc
solve_network	✔	✔	✔
Output	results/<scenario>/networks/base_s_10_elec.nc	same pattern	same pattern

Dunkelflaute modification detail (all stress scenarios):

Modified attribute	Carriers affected	Transformation
generators_t.p_max_pu	onwind, offwind-ac, offwind-dc, offwind-float	× 0.15 on 5 worst days (+ 6-snapshot ramp at edges)
generators_t.p_max_pu	solar, solar-hsat	× 0.10 on 5 worst days (+ 6-snapshot ramp at edges)
All other components	—	Unchanged

Nuclear modification detail (SMR / MSR / LFR scenarios):

Added attribute	Value
New Generator components	3–5 per scenario (see Section 6.5)
p_nom (initial)	0 MW
p_nom_extendable	True
p_nom_max	1,500 MW per site
capital_cost	From processed 2050 costs (18–22 kEUR/MW/a)
marginal_cost	10.6–12.1 EUR/MWh
p_max_pu	0.90 (constant availability)
p_min_pu	0.30
ramp_limit_up/down	0.50 per hour

6.8 Cross-scenario findings comparison table

All values for the **2-week simulation period** (336 hours, 112 snapshots at 3h resolution).
Δ columns show change relative to **base**. Δ vs **dunkelflaute** shows incremental effect of adding nuclear.

6.8.1 System-level KPIs

KPI	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Load (TWh)	21.15	21.15	21.15	21.15	21.15
Generation (TWh)	21.21	21.21	21.21	21.21	21.21
OPEX (M EUR)	269.7	338.8	338.8	338.9	338.8
Objective (M EUR)	269.7	338.8	338.8	338.9	338.8
CAPEX (M EUR)	1,658.9	1,658.9	1,658.9	1,658.9	1,658.9
CO ₂ (kt)	3,283	3,849	3,849	3,849	3,849

KPI	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Solver time (s)	~306	~375–382	~382	~382	~379
Δ OPEX vs base (M EUR)	—	+69.1	+69.1	+69.2	+69.1
Δ OPEX vs base (%)	—	+25.6%	+25.6%	+25.6%	+25.6%
Δ CO₂ vs base (kt)	—	+566	+566	+566	+566
Δ vs dunkelflaute (OPEX)	—	—	–0.001 M EUR	+0.019 M EUR	+0.007 M EUR

6.8.2 Generation mix (TWh, 2-week period)

Carrier	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr	Δ base→dunkelflaute
CCGT	0.34	1.18	1.18	1.18	1.18	+0.84 (+247%)
Coal + lignite	8.74	9.61	9.61	9.61	9.61	+0.87 (+10%)
Wind (all)	8.86	7.21	7.21	7.21	7.21	–1.65 (–19%)
Solar (all)	0.70	0.44	0.44	0.44	0.44	–0.26 (–37%)
Biomass	6.15	6.15	6.15	6.15	6.15	–0
OCGT	0.0003	0.0002	0.0001	0.0009	0.0004	–0
SMR	—	—	–0	—	—	—
MSR	—	—	—	~0	—	—
LFR	—	—	—	—	~0	—

6.8.3 VRE capacity factors (2-week average)

Carrier	base	dunkelflaute	All stress + nuclear scenarios	Δ base→stress
Onshore wind	9.3%	7.5%	7.5% (unchanged)	–18.7%
Offshore wind (AC)	17.8%	14.5%	14.5% (unchanged)	–18.3%
Solar PV	1.4%	0.9%	0.9% (unchanged)	–37.0%

VRE capacity factors are identical across all stress scenarios because Dunkelflaute parameters are the same; nuclear availability does not feed back into VRE profiles.

6.8.4 New build outcomes (optimal capacity)

Technology	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Onshore wind (GW)	73.33	73.33	73.33	73.33	73.33
Solar (GW)	48.77	48.77	48.77	48.77	48.77
CCGT (GW)	30.78	30.78	30.78	30.78	30.78
Coal (GW)	20.35	20.35	20.35	20.35	20.35
Lignite (GW)	19.46	19.46	19.46	19.46	19.46
Battery (MW)	0.07	0.03	0.02	0.14	0.06
SMR (MW)	—	—	0.0007	—	—
MSR (MW)	—	—	—	0.0023	—
LFR (MW)	—	—	—	—	0.0012
H ₂ storage (MW)	~0.008	~0.003	~0.002	~0.015	~0.007

Nuclear capacities are numerical dust (sub-kW); effectively **zero build** in all three nuclear scenarios.

6.8.5 CO₂ emissions breakdown (kt, 2-week period)

Emitter	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr	Δ base→dunkelflaute
Coal	1,885	1,910	1,910	1,910	1,910	+25
Lignite	1,275	1,597	1,597	1,597	1,597	+322
CCGT	67	234	234	234	234	+167
Waste	55	106	106	106	106	+51

Emitter	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr	Δ base→dunkelflaute
Total	3,283	3,849	3,849	3,849	3,849	+566 (+17%)
CO ₂ limit (period)	19,178	19,178	19,178	19,178	19,178	Not binding

6.9 Key findings by scenario — summary matrix

Finding	base	dunkelflaute	dunkelflaute-smr	dunkelflaute-msr	dunkelflaute-lfr
Dunkelflaute stress applied	No	Yes	Yes	Yes	Yes
New nuclear option available	No	No	SMR (5 sites)	MSR (3 sites)	LFR (3 sites)
Nuclear built	—	—	~0 MW	~0 MW	~0 MW
Primary flexibility response	CCGT baseload	CCGT +247%	CCGT (same)	CCGT (same)	CCGT (same)
VRE share of generation	~45%	~36%	~36%	~36%	~36%
Fossil + biomass share	~70%	~78%	~78%	~78%	~78%
CO ₂ limit binding	No	No	No	No	No
Storage utilisation	Negligible	Negligible	Negligible	Negligible	Negligible
Cost impact vs base	Reference	+25.6% OPEX	+25.6% (no delta vs dunkelflaute)	+25.6%	+25.6%
Engineering conclusion	Winter baseline	Gas fills VRE gap	SMR not competitive at placeholder costs	MSR lowest marginal cost but still no build	LFR intermediate, no build

Central conclusion from cross-scenario comparison: The only scenario change that materially affects system outcomes is **Dunkelflaute stress** (base → dunkelflaute). Adding SMR, MSR, or LFR on top of Dunkelflaute produces **no meaningful difference** in dispatch, cost, or emissions under current assumptions — because the optimiser builds effectively zero nuclear capacity.

7. Preliminary Results — Quantitative Analysis

7.1 Summary table (2-week simulation period — correct scaling)

Energy and costs integrated over the 336-hour simulation window with proper snapshot weights:

Scenario	Load (TWh)	Generation (TWh)	OPEX (M EUR)	Objective (M EUR)	CO ₂ (kt)
base	21.15	21.21	269.7	269.7	3,283
dunkelflaute	21.15	21.21	338.8	338.8	3,849
dunkelflaute-smr	21.15	21.21	338.8	338.8	3,849
dunkelflaute-msr	21.15	21.21	338.8	338.8	3,849
dunkelflaute-lfr	21.15	21.21	338.8	338.8	3,849

Note: CAPEX (~1,659 M EUR) is similar across scenarios because it reflects **annuitised cost of the entire installed fleet**, not incremental investment in this 2-week run. Incremental CAPEX from new nuclear is negligible (~0 MW built).

7.2 Generation mix shift — base vs Dunkelflaute (TWh, 2-week period)

Carrier	base	dunkelflaute	Δ	Δ (%)
CCGT	0.34	1.18	+0.84	+247%
Coal + lignite	8.74	9.61	+0.87	+10%
Wind (all)	8.86	7.21	-1.65	-19%
Solar (all)	0.70	0.44	-0.26	-37%
Biomass	6.15	6.15	~0	0%
OCGT	0.0003	0.0002	~0	—
Nuclear (SMR/MSR/LFR)	0	~10 ⁻⁷	~0	—

7.3 Installed capacity (GW) — optimal expansion

Capacities are essentially **identical across all five scenarios** because the 2-week window does not justify large structural changes:

Carrier	Capacity (GW)	Extendable?
Onshore wind	73.3	Yes
Solar PV	48.8	Yes
CCGT	30.8	Yes
Coal	20.4	No (fixed)
Lignite	19.5	No (fixed)
Offshore wind (AC)	11.2	Yes
Biomass	8.0	No
OCGT	6.1	Yes
SMR / MSR / LFR	~10 ⁻⁶	Yes (but builds nothing)
Battery	~10 ⁻⁴	Yes (negligible)

The model's 2024-estimated renewable fleet (~122 GW wind + solar) is retained; the optimiser does not expand VRE within this short window because existing capacity is sufficient except during Dunkelflaute hours.

7.4 VRE capacity factors (2-week average)

Carrier	base CF	dunkelflaute CF	Change
Onshore wind	9.3%	7.5%	~18.7%
Offshore wind (AC)	17.8%	14.5%	~18.3%
Solar PV	1.4%	0.9%	~37.0%

Winter capacity factors are low in absolute terms (January); the Dunkelflaute derating further suppresses them during the 5 worst days.

7.5 Dispatch on worst VRE day (2021-01-25) — Dunkelflaute-SMR scenario

Carrier	Generation (GWh/day)
Coal	163
Lignite	156
CCGT	100
Biomass	64
Waste	22
Onshore wind	7.7
Offshore wind	4.0
Solar	1.6
SMR	~0.000004

On the worst day, **~95% of electricity comes from fossil and biomass**; VRE contributes ~13 GWh (~6% of daily energy).

7.6 Nuclear build detail

Scenario	Total nuclear built (MW)	Generators	Generation (TWh)
dunkelflaute-smr	0.00066	5 × ~0.00013 MW	1.5×10 ⁻⁷
dunkelflaute-msr	0.00225	5 × ~0.00045 MW	5.0×10 ⁻⁷
dunkelflaute-lfr	0.00115	5 × ~0.00023 MW	2.6×10 ⁻⁷

These are **numerical artefacts** (sub-kW capacities), not meaningful engineering outcomes. The LP assigns tiny capacities to avoid degeneracy; they have zero practical significance.

7.7 Comparison script caveat

`compare_scenarios.py` annualises supply by multiplying by `8760 / n_snapshots`, producing values like "911,266 TWh" — **physically meaningless**. The script's plotting function also fails on this data (`ValueError: If using all scalar values, you must pass an index`). **Use period-correct energy (Section 7.1–7.2) for interpretation.** Fix planned in next steps.

8. Energy Engineering Interpretation

8.1 What the Dunkelflaute scenario reveals

The Jan 2021 window captures a **winter high-pressure situation** typical of Central European Dunkelflaute:

1. **Low solar** — short days, low elevation angle; 1–2% capacity factor is realistic for January.
2. **Moderate but insufficient wind** — even before derating, 9–18% CF cannot cover ~63 GW average demand with ~122 GW installed VRE (need ~50%+ CF or storage/interconnection).
3. **Under stress**, the gap is filled by **flexible gas (CCGT)** first because:
 - Marginal cost ~43 EUR/MWh vs coal/lignite must-run baseload.
 - CCGT is extendable and already has 31 GW installed.
 - Ramp limits allow daily cycling.
4. **Coal/lignite increase modestly (+10%)** because they operate near minimum stable output already as baseload; they provide inertia and low-marginal-cost energy but cannot ramp fast enough to cover all peaks.

Engineering conclusion: In the current model setup, **Dunkelflaute is a gas-and-coal event**, not a storage or nuclear event. This aligns with real-world observations from Jan 2021 when Germany relied heavily on conventional generation and imports (imports not modelled here).

8.2 Why nuclear did not build

Three reinforcing reasons:

1. **Capital recovery period vs snapshot window:** Annuitised CAPEX ~18,000 EUR/MW/year must be recovered from dispatch savings over **336 hours**. Even if nuclear displaced CCGT at 43 EUR/MWh every hour, revenue $\approx 43 \times 336 \approx 14,500$ EUR/MW — less than one year's capital cost alone.
2. **High absolute CAPEX:** At 4,500 EUR/kW (SMR), a 1 GW plant costs ~4.5 bn EUR overnight → ~18 kEUR/MW/year annuitised. Competing with CCGT at ~4.8 kEUR/MW/year annuitised CAPEX, gas wins on short horizons.
3. **No carbon price binding:** With CO₂ cap at 17–20% utilisation, coal/lignite remain cheap. Nuclear's zero-carbon advantage is not monetised.
4. **Baseload mismatch:** Nuclear with `p_min_pu = 0.3` wants to run continuously; a 2-week winter window with moderate total demand does not create enough **scarcity pricing** to justify 24/7 firm capacity investment.

Engineering conclusion: The model correctly rejects nuclear under current cost assumptions and short optimisation horizon. This is **not evidence that nuclear is useless for Dunkelflaute** — it indicates the **model configuration is not yet suited to capacity adequacy decisions**.

8.3 Technology ranking (if nuclear were forced to build)

At placeholder costs, marginal cost ranking for dispatch:

`MSR (10.6) < LFR (11.4) < SMR (12.1) < CCGT (43.3)` EUR/MWh

MSR has lowest operating cost but highest CAPEX (5,200 EUR/kW). The optimiser's build decision depends on **utilisation hours** — classic baseload economics.

8.4 Storage and hydrogen

Battery optimal capacity: **~0.07 MW (base), ~0.03 MW (Dunkelflaute)** — effectively zero.

Why: With 2-week horizon, arbitrage between high-wind and low-wind hours within the window does not justify battery CAPEX. H₂ chain similarly unused. Over a full year with repeated Dunkelflaute events, seasonal storage economics would differ substantially.

8.5 CO₂ and policy

Emitting 3.3 Mt in 2 weeks → **~86 Mt/year** if extrapolated linearly (rough, not rigorous). The 500 Mt/year cap allows this easily. For a 2050 net-zero scenario, the cap should be **< 50 Mt/year** or coal/lignite must be retired via `powerplants_filter`.

Top emitters (base, kt over 2 weeks):

Carrier	CO ₂ (kt)	Share
Coal	1,885	57%
Lignite	1,275	39%
CCGT	67	2%

9. Known Limitations & Model Artefacts

Limitation	Severity	Description
Short snapshot window	Critical	Investment decisions not representative of annual/full-life economics
Non-binding CO ₂ cap	High	Coal/lignite fleet inconsistent with 2050 policy narrative
National uniform Dunkelflaute	Medium	Real events have spatial structure
10-cluster aggregation	Medium	Internal congestion within zones ignored
No imports/exports	Medium	Germany relied on neighbours during 2021 event

Limitation	Severity	Description
Placeholder nuclear costs	High	±50% cost swing would change build decisions
No national nuclear cap	Medium	Unrealistic if 5×1500 MW all build in future configs
compare_scenarios annualisation bug	Low (reporting)	Misleading TWh figures in CSV
HiGHS "Unknown" status	Low	Solution usable; gap ~0.003% of objective
Linear programming (no UC)	Medium	CCGT can fractionally dispatch without start costs
Coal still in 2050 fleet	High	<code>powerplants_filter</code> keeps pre-2025 plants without phase-out schedule

10. How to Improve & Change Assumptions

10.1 Dunkelflaute parameters

Parameter	Current	Suggested range	How to change
<code>wind_factor</code>	0.15	0.05 – 0.30	<code>data/inre/dunkelflaute.yaml</code>
<code>solar_factor</code>	0.10	0.02 – 0.20	Same
<code>auto_worst_days</code>	5	3 – 14	Same; or set <code>null</code> and use fixed <code>time_start/time_end</code>
<code>ramp_hours</code>	6	0 – 12	Same
Spatial pattern	Uniform	Zone-varying factors	Extend <code>apply_dunkelflaute.py</code>

Calibration approach: Compare derated profiles against ENTSO-E actual wind/solar generation during Jan 24 – Feb 8, 2021. Compute the ratio of actual to potential generation on each day; use the 5th percentile as `wind_factor/solar_factor`.

10.2 Nuclear techno-economics

Replace `data/inre/custom_costs_nuclear.csv` with literature values:

Source type	Examples
SMR	NEA/IEA "Projected Costs of Generating Electricity", NuScale/ Rolls-Royce SMR studies
MSR	EU SAMOFAR, Moltex, Terrestrial Energy public estimates
LFR	MYRRHA, BREST-OD-300 conceptual designs

Recommended sensitivity band: **±40% on overnight capital cost**, **±10% on availability**, **±5% on efficiency**.

Also consider adding to `add_nuclear_technologies.py`:

- `GlobalConstraint` on total nuclear capacity (national cap, e.g. 10 GW).
- Scheduled `DateIn` enforcement (plants unavailable before 2045).
- Co-location with industrial heat offtake (future sector coupling).

10.3 Policy & fleet

Change	Config key	Suggested value
Binding CO ₂ cap	<code>electricity.co2limit</code>	50e+6 (50 Mt/year) for 2050 net-zero path
Coal phase-out	<code>electricity.powerplants_filter</code>	Exclude coal/lignite: <code>Fueltype != 'Hard coal' and Fueltype != 'Lignite'</code>
Carbon price	PyPSA <code>EmissionPrice</code> constraint	Alternative to hard cap
Renewable expansion limit	<code>estimate_renewable_capacities.expansion_limit</code>	<code>true</code> with cap if matching real pipeline

10.4 Temporal & spatial resolution

Goal	Change	Trade-off
Better investment signal	Extend to full year or 4× seasonal weeks	Solve time ↑ (hours)
Faster iteration	<code>resolution_elec</code> : 24h or 6h	Lose ramp detail
More spatial detail	<code>clusters</code> : [20] or [37]	Memory and solve time ↑
Adequacy focus	Switch to <code>solve_operations_network</code> with fixed capacities	Separates expansion from operation

10.5 Model formulation upgrades

1. **Myopic or perfect foresight expansion** over multiple investment periods (`foresight`: `myopic`).

2. **Unit commitment** for CCGT/OCGT (requires MILP, e.g. Gurobi).

3. **Import/export** links to FR, NL, PL, CZ, AT, DK (PyPSA-Eur multi-country config).

4. **Demand-side response** as shiftable load.

5. Fix `compare_scenarios.py` to report period energy and optional annual extrapolation with explicit scaling factor.

11. Recommended Next Steps (Prioritised)

Phase A — Fix & validate (1–2 days)

- ☐ Fix `compare_scenarios.py` — period-correct energy, working plots, delta table vs base.

☐ Add coal/lignite phase-out scenario — rerun Dunkelflaute without fossil baseload.

☐ Tighten CO₂ cap to binding level (e.g. 50 Mt/year); observe dispatch shift.

☐ Document solver runtime vs target (currently ~5–6 min , within 15 min goal).

Phase B — Nuclear relevance (3–5 days)

- ☐ Literature review → update `custom_costs_nuclear.csv` with cited values.

☐ Operations-only mode: Fix VRE + storage + gas capacities at 2030/2040/2050 levels; optimise dispatch and **marginal value of nuclear** during Dunkelflaute only.

☐ Add national nuclear cap (`GlobalConstraint`, e.g. 5–15 GW).

☐ Sensitivity sweep: wind_factor × solar_factor × nuclear CAPEX (3×3×3 = 27 runs with shared resources).

Phase C — Temporal realism (1–2 weeks)

- ☐ Extend snapshot window to full year 2021 (hourly or 3h) — requires longer solves or cluster reduction.

☐ Multi-period investment with myopic foresight (2030, 2040, 2050).

☐ Calibrate Dunkelflaute against ENTSO-E actuals for Jan–Feb 2021.

☐ Alternative events: Dec 2016, Feb 2012 for robustness.

Phase D — Reporting & policy (ongoing)

- ☐ Cross-border flows and EU context.

☐ Levelised cost of lost load (LOLP) / unserved energy metric during stress days.

☐ Publication-ready figures: duration curve, dispatch stack for worst day, price duration.

☐ Peer review of assumptions document.

12. Appendix: File Reference & Commands

12.1 Reproduce results

```
cd /path/to/INRE-Project-PyPSA-EUR
pixi install && pixi shell

# All five scenarios
snakemake --call solve_elec_networks \
  --configfile config/inre/config.base.yaml \
  --configfile config/inre/config.scenarios.yaml

# Comparison (after fixing script)
python scripts/inre/compare_scenarios.py --output-dir results/inre-comparison
```

12.2 Key result files

Path	Content
<code>results/base/networks/base_s10_elec_.nc</code>	Solved base case network
<code>results/dunkelflaute*/networks/*.nc</code>	Stress and nuclear variants
<code>results/inre-comparison/comparison_table.csv</code>	KPI summary (note scaling caveat)
<code>results/*/logs/apply_inre_network/*.log</code>	Dunkelflaute day selection log
<code>results/*/logs/solve_network/*_solver.log</code>	HiGHS solver output

12.3 Glossary

Term	Definition
Dunkelflaute	"Dark doldrums" — extended period of low wind and solar output
SMR	Small Modular Reactor (~50–300 MW modules)
MSR	Molten Salt Reactor (liquid fuel/coolant)

Term	Definition
LFR	Lead-cooled Fast Reactor (fast neutron spectrum)
p_max_pu	Per-unit availability factor (0–1) for renewable generators
CAPEX / OPEX	Capital / operating expenditure
CF	Capacity factor = energy output / (capacity × time)

Document History

Version	Date	Notes
1.0	2026-06-22	Initial progress report after first full 5-scenario Phase 2 run

This report reflects the state of the INRE project as of the first successful multi-scenario solve. All numerical results should be treated as preliminary until assumptions P1–P3 (CO₂, coal fleet, snapshot window) are revised for 2050 policy consistency.